Herahakatha — Logsheet Web App Plan

Oct 15, 2026 · @Shirhaan

Summary

Herahakatha replaces the powerhouse Excel logsheets with one PHP + MySQL web app where operators enter each hourly reading on their phone at the meter, and managers open the same web app on a PC for reports and trends. It covers the four logs in use today: hourly feeder readings (26 feeders), half-hourly solar output, daily meter readings for water, solar inverters and generators, and the hourly generator set log.

The main win is not just a nicer form. Today one typo like `11..21` silently drops out of a day's total, sheets carry the wrong date, and one monthly total counts only 4 of 9 solar inverters. Herahakatha checks every reading as it is typed, computes all consumption and totals itself, and keeps every reading in one place, so a month or a year can be compared in seconds.

The build runs in four phases: data entry first, then reports, then importing the 2025–2026 history, then analysis and alerts.

What the logsheets record today

There are five kinds of reading, and they fall into two patterns: **interval logs** (on each hour, or half hour for solar, an operator goes to the meters and notes the instantaneous kW at that moment) and **cumulative meters** (a running counter read once a day, where consumption = today's reading minus yesterday's).

Log	File today	Frequency	What is recorded	Unit
Feeder reading 1	`Feeder Reading 1.xlsx`, one sheet per day	Hourly, 00:00–23:00	HV Network (main incomer) + 11 feeders: JSB-9001, JSB-8001, JSB-9002, JSB-8002, Ex. Staff Blk, S. Staff Blk, MDB 116-117 BV 169-189, MDB 118-119 BV 272-290, Katheeb Villa, STP Eng Office, Laundry Irons	kW

Log	File today	Frequency	What is recorded	Unit
Feeder reading 2	`Feeder Reading 2.xlsx`, one sheet per day	Hourly, 00:00–23:00	14 feeders: Main Store, Main Kitchen, Security Hut, Heat Ex. Pump, MDB 120-121 BV 150-167, IT Bldg, HR / Mosque, Power House, Laundry Machines, Laundry Dryers, Water Plant, Incinerator Workshop, Pump Room, Carpentry	kW
Solar log	`solar log <month>.xlsx` — blank "page" sheets for handwriting, then a "Final" sheet	Every 30 min, 06:00–18:00	Total plant kW from the inverter display, one value per slot (25 slots a day)	kW
Water & Energy	`01.Water & Energy <month>.xlsx` — Water, Solar, Electricity, Statistics sheets	Daily	Water meter Lines 1–11 (only Line 1 Main in use); Solar Inverters 1–9 kWh counters; Generators 1–4 kWh counters; powerhouse diesel litres per day from a flow meter	m³, kWh, L
Generator set log	`GENERATOR SET LOG SHEET.xlsx` (blank template)	Hourly while running, 08:00–07:00	Per genset: voltage, frequency, load current R/Y/B, power factor, active load, lube oil pressure, water temp, energy, running hours, battery volts, RPM, visual check, solar active power	V, Hz, A, kW, bar, °C, kWh, h

The Statistics sheet is the monthly report: daily water (m³), daily electricity (kWh), powerhouse diesel (litres) and **kWh per litre**, the fuel-efficiency figure for the generators. In August 2026 that averaged 13,668 kWh and 3,834 L a day.

The feeder layouts have not changed since at least August 2025, and an earlier export (`meter_readings_bulk1/2.sql`) already numbers them as panels 6–17 and 18–31. Herahakatha can keep those IDs so that export loads straight in.

Problems found in the current sheets

The biggest reporting error is the solar total: the August 2026 sheet sums only Inverters 1–4, so 1 Aug shows 1,000 kWh when all 9 inverters produced 2,062 kWh. Every other issue below is a data-entry slip that Excel accepts silently.

Problem	Example found	Effect	Fix in Herahakatha
Solar daily total misses inverters	`=SUM(D10:D13)` covers Inverters 1–4 of 9	Solar under-reported by about half	Totals computed from all active meters, never a hand-made range
Typos stored as text	`11..21`, `23.8/4`, `7.72.`, `8.4710.51`, `2\5.84` — 18 in Sep 2026, 14 in Aug 2025	The cell drops out of the day's SUM, no warning	Numeric keypad + reject anything that is not a number
Decimal point missed	Solar 1 Sep: `3.54` and `5.38` between values of 239–497	A dip that never happened	Warn when a value is far off the slot's usual range or neighbours
Wrong date on the sheet	Sheet `01.09.2026` dated 2026-09-10; sheets `04.07.2026` – `31.07.2026` inside the September file	Readings filed under the wrong day	Date picked once per session; one record per meter per timestamp
Hand-typed consumption	Generator kWh typed, not computed; one day off by 4.5%	Totals cannot be trusted	Consumption always = reading − previous reading × multiplier
Hidden meter multiplier	Generator counters read ÷10 to get kWh (34,165,000 − 34,147,000 = 18,000 → 1,800 kWh)	Easy to apply wrongly	Multiplier stored per meter in settings

Problem	Example found	Effect	Fix in Herahakatha
Mismatched file and data	`Water & Energy AUG 2026` has Water and Electricity dated July	Confusing month boundaries	Reports generated from dates, not file names
Missing totals	Water total row blank; "Other diesel" average shows `#DIV/0!`	Gaps in the monthly report	Report layout fixed and tested once

Users and roles

Four roles keep the app simple: operators only see entry screens, and only admins change meters or settings.

Role	Who	Can do
Operator	Powerhouse shift staff, solar/water readers	Enter and correct readings for the current and previous day; see today's progress
Supervisor	Shift-in-charge, chief engineer	Everything an operator can, plus edit any past reading (with a reason), approve a day, view reports
Viewer	GM, finance, sustainability	Reports, dashboards and exports only
Admin	IT (Shirhaan)	Users, meters, feeders, multipliers, limits, imports, backups

Every reading stores who entered it and when; every edit after saving keeps the old value, new value, user and reason in an audit log.

Modules and screens

The app opens on a **Today** screen that shows what is still missing, so an operator never has to hunt for the right sheet. On a phone, navigation is a bottom bar (Today · Enter · Reports · More); on a PC it becomes a left sidebar with the same items. It is one web app at one address: the same pages rearrange for the screen size, so there is no separate mobile app to install or maintain.

Today (home). One card per log with progress and the next reading due: "Feeders — 14:00 due, 13 of 24 hours done", "Solar — 9 of 25 slots", "Daily meters — not done", "Genset 3 running — log 15:00". Missed slots show in amber; tapping a card opens that entry screen

at the right time slot.

Feeder entry (hourly).

- The screen follows the hourly round: on each hour the operator walks the panels and enters each feeder's kW as it is read. The app files the value under that hour (e.g. 14:00) and also records the actual save time.
- Phone: opens on the hour that is due; the 26 feeders appear in walk order, one large numeric field each, with last hour's value greyed beside it. Each value saves the moment it is entered and the cursor moves to the next feeder, so the whole round is done at the panels.
- PC: the same page shows as a grid like today's sheet — 24 hours × feeders, arrow-key movement, paste from Excel — for supervisors checking a day or catching up.
- Feeder 1 and Feeder 2 stay as two tabs so the screens match how the panels are walked.

Solar entry (half-hourly). One field per solar plant for the current slot, 06:00–18:00, copied from the inverter display; the day's curve draws live underneath so a missed decimal stands out at once. A plant added later in Settings appears here automatically.

Daily meters. One form per morning: water Line 1 (other lines hidden until enabled), every solar inverter (new ones included once added), Generators 1–4, and the powerhouse diesel flow meter. Each row shows yesterday's reading and the computed consumption as you type.

Generator set log. Pick a genset, then one hourly row of 15 fields: voltage, Hz, R/Y/B amps, PF, kW, oil pressure, water temp, kWh, running hours, battery V, RPM, visual check (OK / Not OK + note) and solar kW. Only running gensets ask for readings, and each genset is linked to its Generator 1–4 meter so hourly kWh can be checked against the daily reading.

Reports and **Analysis** (sections below), and **Settings** for meters, feeders, groups, multipliers, limits and users.

Design rules for ease of use:

- Numeric keypad on phones (`inputmode="decimal"`), large touch targets, one task per screen.
- Nothing to calculate by hand: consumption, totals, kWh/L are always computed.
- Each value saves the moment it is entered, so an interrupted round loses nothing; values typed during a Wi-Fi drop or an internet outage wait in the browser and send when the connection returns.
- English labels with the feeder names exactly as on the panels.

MySQL data model

One generic `readings` table holds every feeder, solar, water, inverter, generator and diesel value; what a reading means comes from its row in `meters`. Adding a new feeder, a second water line or the new solar plant and its inverters is then a Settings change, not a code change; reports group meters by kind, so new solar is counted in every solar total automatically. The genset log has 15 values per hour, so it gets its own table.

Table	Holds	Key columns
`sites`	Property (Canareef now; room for another hotel later)	name, timezone
`users`	Logins and roles	username, password_hash, role, active
`meter_groups`	How meters are shown: Feeder 1, Feeder 2, Solar, Water, Inverters, Generators, Fuel	name, sort_order
`meters`	Every point that is read	code, name, kind, reading_type (interval / cumulative), unit, multiplier, interval_min, window_start, window_end, min_value, max_value, is_incomer, sort_order, active
`readings`	One value per meter per timestamp	meter_id, reading_at, value, entered_by, entered_at, source (app / import), note — UNIQUE (meter_id, reading_at)
`reading_audit`	Every change after first save	reading_id, old_value, new_value, changed_by, changed_at, reason
`genset_logs`	Hourly generator set log	meter_id, log_at, volts, hz, amp_r, amp_y, amp_b, pf, kw, oil_bar, water_c, kwh, run_hours, battery_v, rpm, visual_ok, visual_note, solar_kw, entered_by
`daily_totals`	Pre-computed per meter per day, refreshed on every save	meter_id, day, total, consumption, avg, peak, peak_at, slots_filled
`day_status`	Supervisor sign-off per log per day	site_id, day, meter_group_id, approved_by, approved_at

The core two tables:

```
CREATE TABLE meters (
  id            INT UNSIGNED PRIMARY KEY,
  site_id       INT UNSIGNED NOT NULL,
  group_id      INT UNSIGNED NOT NULL,
  code          VARCHAR(40)  NOT NULL,          -- e.g. JSB-9001, INV-5, GEN-
2
  name          VARCHAR(100) NOT NULL,
  kind          ENUM('feeder','solar','water','inverter','generator','fuel')
NOT NULL,
  reading_type  ENUM('interval','cumulative') NOT NULL,
  unit          VARCHAR(10)  NOT NULL,          -- kW, kWh, m3, L
  multiplier    DECIMAL(10,4) NOT NULL DEFAULT 1, -- generators: 0.1
  interval_min  SMALLINT     NULL,              -- 60 feeders, 30 solar, NULL
daily
  window_start  TIME NULL, window_end TIME NULL, -- solar 06:00-18:00
  min_value     DECIMAL(14,3) NULL, max_value DECIMAL(14,3) NULL,
  is_incomer    TINYINT(1) NOT NULL DEFAULT 0,  -- HV Network
  sort_order    SMALLINT NOT NULL DEFAULT 0,
  active        TINYINT(1) NOT NULL DEFAULT 1,
  UNIQUE KEY (site_id, code)
) ENGINE=InnoDB;

CREATE TABLE readings (
  id          BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
  meter_id    INT UNSIGNED NOT NULL,
  reading_at  DATETIME NOT NULL,
  value       DECIMAL(14,3) NOT NULL,
  entered_by  INT UNSIGNED NOT NULL,
  entered_at  DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
  source      ENUM('app','import') NOT NULL DEFAULT 'app',
  note        VARCHAR(255) NULL,
  UNIQUE KEY uq_meter_time (meter_id, reading_at),
  KEY idx_time (reading_at),
  FOREIGN KEY (meter_id) REFERENCES meters(id)
) ENGINE=InnoDB;
```

`reading_at` is the slot a reading belongs to (14:00) and `entered_at` is when it was actually saved, which shows whether rounds run on time. Size is not a concern: 26 feeders × 24 h plus solar and daily meters is about 240,000 rows a year. Feeder meter IDs are seeded as 6–31 in the same order as the old `meter_readings` export, so that file imports with a simple column rename.

Validation rules

Hard rules block the save; soft rules show an amber warning and ask "Save anyway?" with an optional note, because unusual readings are sometimes real (a feeder tripped, a genset stopped).

Rule	Applies to	Type
Must be a plain number, one decimal point, no other characters	All	Block
Not negative	All	Block
One reading per meter per time slot; a second entry becomes an edit with audit	All	Block
No future timestamps; operators can only enter today and yesterday	All	Block
Cumulative reading lower than the previous one	Water, inverters, generators	Block (supervisor can override for a meter replacement or rollover)
Outside the meter's min/max set in Settings	All	Warn
More than 3× or less than ⅓ of the same slot's 14-day median (catches `3.54` for `354`)	Feeders, solar	Warn
Sum of feeders far from HV Network for that hour	Feeder hours	Warn on the hour summary
Daily consumption more than 40% off the 7-day average	Daily meters	Warn
Genset oil pressure, water temperature, battery volts, Hz outside limits	Genset log	Warn in red, flagged on the dashboard

A reading saved more than 20 minutes after its slot is kept but marked late, so the completeness report shows when rounds slip. Limits and percentages above are starting values; each is a setting the admin can tune once a few weeks of clean data exist.

Reports and analysis

Every report is built from the stored readings, so the monthly Water & Energy statistics appear automatically at month end with no copying between sheets.

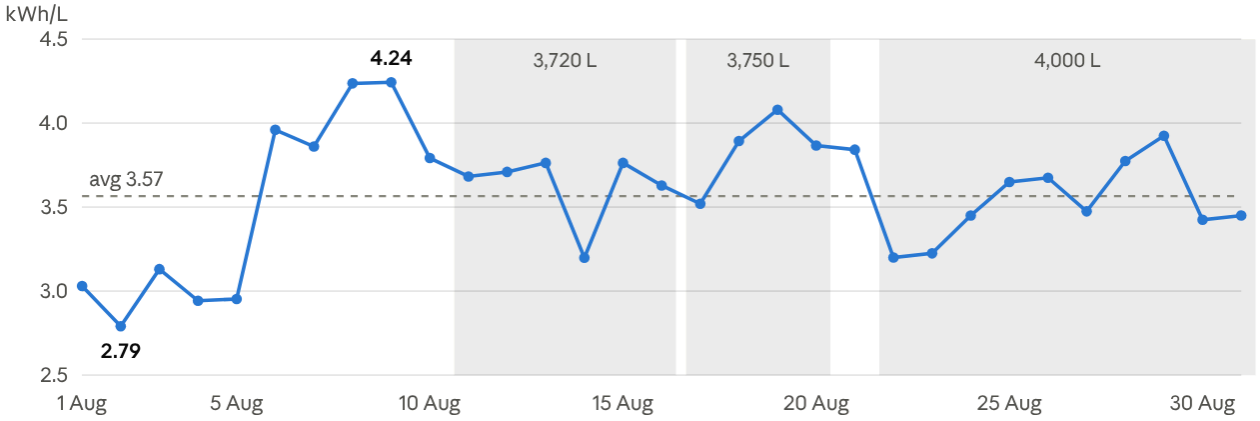
Report	What it shows	Export
Daily feeder report	24 h × 26 feeders, estimated daily kWh per feeder (sum of hourly kW × 1 h), peak hour, HV Network vs sum of feeders	Print/PDF, Excel
Daily solar report	Half-hourly curve, estimated energy (sum of kW × 0.5 h) checked against the inverter kWh counters, peak output, per solar plant	Print/PDF
Monthly Water & Energy statistics	Replaces the Statistics sheet: water m³, generator kWh, solar kWh (every inverter, new plants included), diesel L, kWh/L, solar share %, totals and averages	Print/PDF, Excel
Feeder consumption ranking	kWh per feeder per day/month, share of total, change vs last month	Excel, CSV
Genset report	Running hours, kWh, average load, PF, out-of-limit events per genset	Print/PDF
Data completeness	Missing slots per log per day, warnings saved anyway, who entered what	On screen

Analysis screens:

- **Dashboard** tiles: today's electricity kWh, solar kWh and share, water m³, kWh/L, gensets running, readings missing.
- **Trend explorer**: any meter or group, any range, by hour/day/week/month, with "same period last year".
- **Load profile**: average 24-hour curve per feeder (weekday vs weekend) and a feeder × hour heatmap to spot peak loads.
- **Losses**: HV Network minus the sum of all feeders, per hour and per day.
- **Solar performance**: kWh per inverter against the others, so a failing inverter shows within a day.
- **Fuel efficiency**: kWh per litre over time, as below.

August averaged 3.57 kWh/L, but diesel litres repeat on 20 of 31 days

Daily generator kWh ÷ powerhouse diesel litres, Aug 2026. Shaded: the same litres logged day after day.



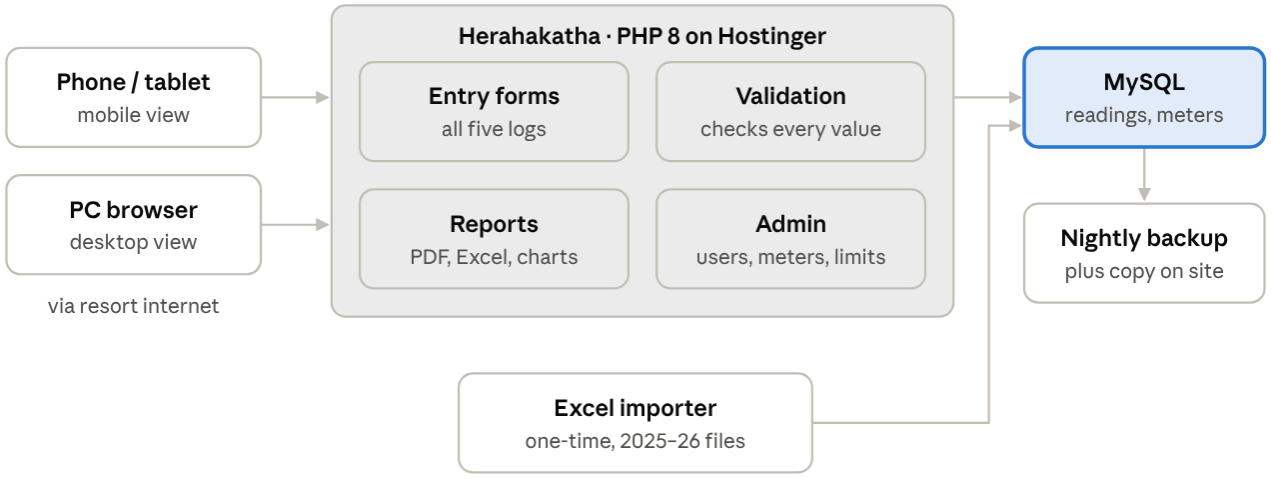
01.Water & Energy AUG 2026.xlsx · Statistics sheet · 31 days

This is the kind of finding the analysis view surfaces: August's daily kWh/L swings from 2.79 to 4.24, but the diesel column holds the same figure for 6, 4 and then 10 days in a row. Those look like estimates. Herahakatha takes the diesel flow meter reading each day and computes litres used from it, so kWh/L becomes a figure the team can act on.

Tech stack and architecture

Herahakatha is one plain PHP 8 application on Apache with MySQL 8, hosted on Hostinger and reached over the resort internet from phones and PCs. Development and testing run locally in Laragon (herahakatha.test), and releases are uploaded to Hostinger.

Phones and PCs use one PHP app; MySQL holds every reading



Herahakatha architecture · one PHP app, one MySQL database

Layer	Choice	Why
Server	Hostinger web hosting: PHP 8.2+ (set in hPanel), its MySQL database, free SSL	Matches existing skills and hosting; no build step
Structure	Lean MVC folders (`public/` , `app/Controllers` , `app/Models` , `views/`), Composer autoload, PDO with prepared statements	Easy to read and maintain without a big framework
UI	Bootstrap 5 (responsive grid, bottom nav on phones), Alpine.js for live form behaviour	Works on phone and PC from one codebase
Charts	Chart.js	Simple, fast, good on mobile
Exports	PhpSpreadsheet (Excel), Dompdf or print stylesheet (PDF)	Same report formats people use now
Phone view	Same web app, responsive layout; service worker queues values while offline	Rounds carry on through Wi-Fi drops and internet outages; queued values send when the link returns
Security	`password_hash` , sessions, CSRF tokens, role checks on every route, HTTPS with Hostinger's SSL, login attempt limits	Readings are operational records
Backup	Nightly `mysqldump` , run by an hPanel cron job, 30 days kept on the account, plus Hostinger's own backups and a weekly copy downloaded to the resort	One copy off-site, one at the resort

Entry forms save through a small JSON API (`POST /api/readings`) so the phone screen stays fast and can retry a save after a dropped connection.

On Hostinger. An hPanel cron job runs nightly for the backup, the missing- and late-reading check, and alert emails. Composer packages (PhpSpreadsheet, Dompdf) are installed over SSH, or uploaded with the `vendor` folder if the plan has no SSH. Because entry now depends on the resort's internet, an outage stops live saving; the offline queue keeps rounds going, and the Today screen shows how many values are waiting to send.

Importing the historical Excel data

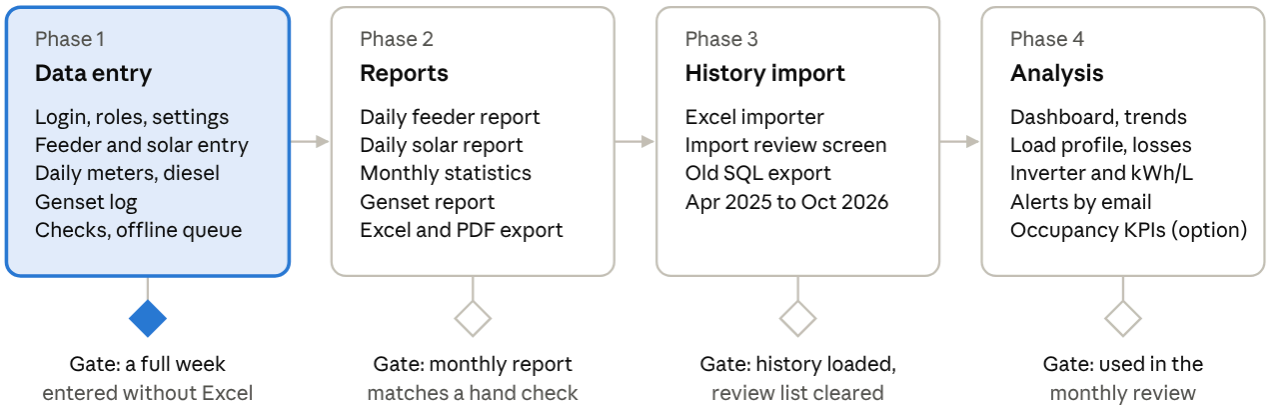
The importer brings in April 2025 to October 2026 so trends and "same month last year" work from day one. It is an admin upload page on the Hostinger site using PhpSpreadsheet. Files go up one month at a time so each run stays inside the hosting time limit, or it runs as a command over SSH if the plan has it. It never trusts a sheet's own date.

1. Read the date from the folder (`Feeder Report/2026/9.sep`) and the day number from the sheet name's first two digits; ignore the date cell, which is often wrong.
2. Map each column header (`JSB-9001` , `MAIN KITCHEN` …) to its meter by name, so a column moved in an older file still lands correctly.
3. Solar: read only the "Final" sheet; skip the blank handwriting pages. Fix cells stored as times by mistake (e.g. a `10:00` slot read as 1900-01-10).
4. Water & Energy: import the raw meter readings and diesel litres only; recompute consumption and totals in the app instead of copying Excel's results.
5. Any cell that is not a clean number goes to an **import review** screen showing file, sheet, cell and the raw text (`11..21`), with a suggested value to accept or correct. Nothing is guessed silently.
6. Skip duplicate folders (`2025/June/May` repeats `2025/May`) and Office temp files (`~$Feeder Reading 1.xlsx` , `Thumbs.db`).
7. Load `meter_readings_bulk1/2.sql` through phpMyAdmin where it overlaps, mapping `panel_id` → `meter_id` .
8. Print a summary per month: rows imported, cells sent to review, days missing.

Build phases

Phase 1 alone replaces the Excel entry; each later phase builds on clean data from the one before.

Four phases: operators stop using Excel after phase 1



Build phases · each ends at a gate before the next starts

History comes before analysis so that trend and year-on-year views have 18 months of data when they first open. During phase 1, staff can run the app and Excel side by side for one week to compare totals.

Decisions and open questions

Shirhaan answered these on 15 Oct 2026 and chose Hostinger for hosting; the questions still open do not block phase 1.

Question	Answer	Effect on the plan
Inverters 5–9 left out of the solar total	Formula mistake	The app totals every inverter; solar figures will read higher than the old monthly sheets
What a feeder value is	Instantaneous kW, read on the hour	Daily feeder energy is an estimate: sum of the 24 hourly kW × 1 h
Solar half-hourly value	Total plant kW from the inverter display	Estimated daily energy (sum × 0.5 h) is cross-checked against the inverter kWh counters
How diesel is measured	Flow meter	Diesel becomes a daily cumulative meter; litres used = today's reading − yesterday's
Is the genset log used	Yes	Genset log stays in phase 1, linked to Generators 1–4

Question	Answer	Effect on the plan
More solar later	Expected	Solar plants and inverters are added in Settings; entry screens and reports pick them up automatically
Mobile	Same web app, mobile view	One responsive web app at one address; nothing to install

Still open:

- [ ] How many inverters will the new solar add, and will it have its own display for the half-hourly kW?
- [ ] Diesel flow meter: read once a day at what time, and does it show cumulative litres?
- [ ] What were panel IDs 1–5 in the old `meter_readings` export, and who is user 4?
- [ ] Which Hostinger plan (SSH access and backup frequency differ by plan), and which domain or subdomain?
- [ ] Who are the first operators and supervisors, and do they share phones or have their own?
- [ ] Should water lines 2–11 and the "Other diesel" column be set up now or left hidden?
- [ ] Later option: pull daily occupancy from Opera to report water and kWh per occupied room?